

Chapter 7: Particulate Nature of Matter

NCERT Class 8 Science (Curiosity)

Q1. Why is it possible to pile up stones or sand, but not a liquid like water?

Answer: Stones and sand grains are solids. The constituent particles inside solids are held together by very strong interparticle forces of attraction and have fixed positions with minimum interparticle space . This allows solid pieces to hold a definite shape and stack on top of one another. In contrast, liquid particles are free to move past each other because their interparticle attractions are weaker. Therefore, water flows and spreads out under gravity instead of piling up .

Q2. Why does water take the shape of folded hands but lose that shape when released?

Answer: Water is a liquid , which has no fixed shape but has a fixed volume. Liquid particles are free to move within a limited space. When held in folded hands, the water particles conform to the shape of the hands . Once the hands are opened, the constraint holding them in place is removed, and the liquid flows away due to gravity and particle movement.

Q3. We cannot see air, so how does it add weight to an inflated balloon?

Answer: Air is matter in a gaseous state. Matter is composed of tiny constituent particles (atoms and molecules) that possess mass. Although individual air particles are invisible to the naked eye , millions of air particles pumped into a balloon collectively possess mass. The accumulation of these particles adds measurable weight to the inflated balloon.

Q4. Is the air we breathe today the same that existed thousands of years ago?

Answer: Yes, at the particulate level, the fundamental constituent particles (molecules of nitrogen, oxygen, carbon dioxide, argon, etc.) are essentially the same . According to the particulate nature of matter, particles are neither created nor destroyed during physical and chemical processes; they simply cycle through the Earth's atmosphere, oceans, and living organisms. However, human activities over centuries have added new suspended particulate matter (SPM) and pollutants to the air .

Q1. Choose the correct option:

The primary difference between solids and liquids is that the constituent particles are:

- (i) closely packed in solids, while they are stationary in liquids.
- (ii) far apart in solids and have fixed position in liquids.
- (iii) always moving in solids and have fixed position in liquids.
- (iv) closely packed in solids and move past each other in liquids.

Correct Answer: (iv) closely packed in solids and move past each other in liquids.

Q2. Which of the following statements are true?

Correct the false statements.

(i) Melting ice into water is an example of the transformation of a solid into a liquid.

- Answer: True

(ii) Melting process involves a decrease in interparticle attractions during the transformation.

- Answer: True

(iii) Solids have a fixed shape and a fixed volume.

- Answer: True

(iv) The interparticle interactions in solids are very strong, and the interparticle spaces are very small.

- Answer: True

(v) When we heat camphor in one corner of a room, the fragrance reaches all corners of the room.

- Answer: True

(vi) On heating, we are adding energy to the camphor, and the energy is released as a smell.

- Answer: False
- Correction: On heating, thermal energy increases the kinetic energy of camphor particles, causing them to convert into gaseous particles. These gaseous camphor particles mix with air particles and move freely across the room; smell is not energy itself, but the sensory response to those airborne particles hitting our olfactory senses.

Q3. Choose the correct answer with justification: If we could remove all the constituent particles from a chair, what would happen?

- (i) Nothing will change.
- (ii) The chair will weigh less due to lost particles.
- (iii) Nothing of the chair will remain.

Correct Answer: (iii) Nothing of the chair will remain.

Justification: Matter is entirely composed of its constituent particles. The chair is simply a structured macroscopic object built out of billions of these microscopic particles. If all constituent particles are removed, there is no matter left, so nothing of the chair will remain.

Q4. Why do gases mix easily, while solids do not?

- **Answer:** 1. **Interparticle Spacing:** Gases have very large interparticle spaces compared to solids, which are tightly packed. 2. **Forces of Attraction:** In gases, interparticle attractions are negligible, allowing particles to move freely in all directions at high speeds. This allows gas particles to easily slip into spaces between other gas particles (diffusion). 3. In solids, particles are fixed in position by very strong attractive forces and can only vibrate in place, preventing them from mixing easily

Q5. When spilled on the table, milk in a glass tumbler flows and spreads out, but the glass tumbler stays in the same shape.

Justify this statement.

- **Answer:**
 - **Milk (Liquid):** Milk is a liquid. Its constituent particles have weaker interparticle forces of attraction and can move freely past one another. Therefore, when spilled, it loses its container shape and flows across the table.
 - **Glass Tumbler (Solid):** The glass tumbler is a solid. Its constituent particles are bound by extremely strong interparticle forces in a fixed pattern, preventing particle movement. Hence, it retains its rigid shape.

Q6. Represent diagrammatically the changes in the arrangement of particles as ice melts and transforms into water vapour.

- **Answer Explanation:**
 1. **Ice (Solid Phase):** Particles are tightly packed in a regular geometric pattern with minimal interparticle space and vibrate about fixed positions.
 2. **Water (Liquid Phase):** On heating, particles gain thermal energy, breaking fixed positions. Particles are slightly farther apart, organized randomly, and can slide past each other.
 3. **Water Vapour (Gas Phase):** On further heating to the boiling point, particles overcome attractive forces completely. Particles become widely separated and move rapidly and randomly in all directions.

Q7. Draw a picture representing particles in the following:

(i) **Aluminium foil (Solid):**

- **Representation:** Draw tightly packed spheres arranged in neat, orderly rows and columns with minimal gap between them.

(ii) Glycerin (Liquid):

- **Representation:** Draw spheres packed somewhat loosely without fixed rows, with small spaces between them.

(iii) Methane gas (Gas):

- **Representation:** Draw very few spheres widely spaced apart from each other inside a closed box, with arrows showing free movement in all directions.

Q8. Observe Fig. 7.16a which shows the image of a candle that was just extinguished after burning for some time. Identify the different states of wax in the figure and match them with Fig. 7.16b showing the arrangement of particles.

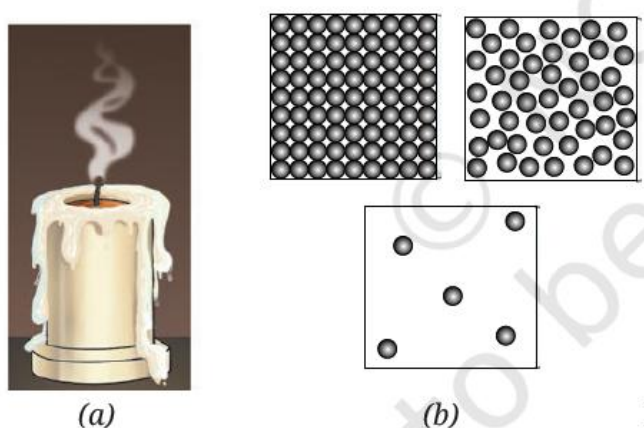


Fig. 7.16

Answer: Solid Wax (Unburnt candle body): Matches the diagram showing tightly packed, ordered particles.

Molten/Liquid Wax (Pool near the wick): Matches the diagram showing slightly loosely packed particles capable of sliding.

Wax Vapour/Gas (Smoke rising from extinguished wick): Matches the diagram showing particles widely separated from each other.

Q9. Why does the water in the ocean taste salty, even though the salt is not visible?

Explain.

- **Answer:**
 - When salt dissolves in ocean water, its constituent particles separate from each other.
 - These salt particles are extremely tiny—so small that they cannot be seen even under a microscope .
 - They slip into the tiny **interparticle spaces** present between the water particles.
 - Because the salt particles are uniformly distributed throughout the solution, they cannot be seen visually, but their presence is detected by our taste buds.

Q10. Grains of rice and rice flour take the shape of the container when placed in different jars. Are they solids or liquids?

Explain.

- **Answer:** They are **Solids**.
- **Explanation:**
 - Each individual grain of rice (or individual particle of rice flour) has a **fixed shape and a fixed volume** of its own.
 - When poured into a jar, the overall collection of grains fills the container like a fluid because the tiny independent solid pieces can pile on top of each other.
 - Since the individual grain retains its shape regardless of the container, rice and rice flour are classified as solids.